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Experimental and numerical investigations on the water vapor adsorption isotherms and kinetics of binderless zeolite 13X

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ABSTRACT

Sorption materials such as zeolite or silica gel are intensively investigated for thermochemical energy storage applications. Information on the adsorption characteristics of these materials are of utmost importance when designing a process for thermochemical energy storage.

For binderless zeolite 13X beads the water vapor adsorption isotherms and the adsorption kinetics have been studied in detail by experimental investigation in a sorption analyzer (adsorption isotherms) and fixed bed reactor (adsorption kinetics). The experimental investigations have shown that this zeolite is characterized by a high water uptake and very fast reaction kinetics.

Both properties are of great importance when using this material for thermochemical energy storage applications. A high water uptake is required for a high energy storage density whereas high reaction kinetics are essential for a high thermal power output during adsorption.

For an open sorption process, a numerical model of the water vapor adsorption of binderless zeolite 13X beads has been developed and validated by experimental investigations. This model allows an in-depth understanding of the heat and mass transfer during the adsorption process and can furthermore be used for the designing process of thermochemical energy store.

1. Introduction

For an efficient use of fluctuating, renewable energy sources, e.g. solar energy, energy storage systems are of utmost importance. They will enhance energy supply security and improve energy efficiency in all areas of energy conversion, distribution and energy use.

Almost 50% of the European final energy consumption is used for heating and cooling. The building sector alone accounts for more than 35% of the final energy consumption, of which 75% is needed for domestic hot water preparation and room heating [1]. Therefore, an efficient and economic heat production is important to cut the consumption of fossil energy demand in this sector.

Solar thermal systems are an excellent technology to decrease the fossil energy demand in the building sector. Today's typical solar thermal systems are designed as solar assisted heating systems with solar fractions of about 20% to 30%. The remaining heat demand is covered by a conventional fossil fuelled backup heater. In order to significantly enhance the solar energy share in the building sector solar thermal systems with efficient and compact long-term thermal energy stores are needed to store solar energy from summer to winter.

Thermochemical energy storage (TCES) represents an important step towards a new generation of heat storage technology due to their high energy storage density and their almost loss-free storage concept.

In thermochemical energy stores energy is stored and released by a reversible chemical and/or physical reaction. In low temperature applications where the heat is required at temperatures below 100 °C water vapor adsorption on sorption materials (e.g. zeolite or silica gel) and hydration reactions of hygroscopic salts (e.g. magnesium sulphate or magnesium chloride) are the most studied processes for TCES ([2–6]). Eq. (1) describes the basic principle of the charging and discharging process of a thermochemical energy store:

$$A_{(s)} + x \cdot \mathrm{H_2O}_{(g)} \rightleftarrows \mathrm{A} \cdot x \cdot \mathrm{H_2O}_{(s)} + \Delta H \tag{1}$$

The TCES is charged by bringing the solid reactant A in contact with water vapor ($H_2O_{(g)}$). The solid takes up the water to form the product $A{\cdot}x{\cdot}H_2O$ and heat of adsorption or hydration (ΔH) is released. This process is reversible. When heat is supplied (charging of the TCES), the product $A{\cdot}x{\cdot}H_2O$ is decomposed into its elements

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Nomenclature

A	adsorption potential
c_p	heat capacity
D	dispersion coefficient
d_s	particle diameter
E	characteristic energy
f_1	d'Arcy factor
f_2	Forchheimer factor
ΔH	molar enthalpy, molar heat of adsorption
Δh	specific enthalpy, specific heat of adsorption
h	heat transfer coefficient
k_{LDF}	mass transport resistance
n	heterogeneity parameter (Dubinin-Astakhov equation)
p	pressure
p_0	saturation pressure
p_w	partial water vapor pressure
R_w	specific gas constant of water vapor
R_i, R_a	inner and outer diameter of the reactor wall
T	temperature
u	fluid velocity
W	adsorption volume
W_0	maximum adsorption volume
X^*	adsorption equilibrium
x_s	water loading
x_w	water content of the air
z, r	coordinates

Greek symbols

β	coefficient of thermal expansion
δ	diffusion coefficient
ε	porosity
η	viscosity
Λ	effective thermal conductivity
λ	thermal conductivity
μ	tortuosity
ρ	density

Subscripts

20C	at 20 °C
ads	adsorbed
ax	axial
eff	effective
f	fluid
M	macro-pore
R	reactor
rad	radial
s	solid
w	water
∞	infinity
Re_0	particle Reynolds number

water vapor ($H_2O_{(g)}$) and the reactant *A*. A quasi loss-free storage of the energy over long periods of time is possible if the reactant *A* is stored separately from water vapor.

For a high energy storage density a high water uptake of the material and a high heat of adsorption/hydration is required. In order to obtain a high thermal power output during the discharging of the store fast reaction kinetics of the material are desirable.

A central element of the thermochemical energy store is the reactor, where the charging and discharging of the thermochemical energy store take place. Innovative reactor designs adapted to the specific application and storage material are required to allow these processes to be carried out with high efficiency. For the design process of the reactor detailed information on the storage material characteristics, especially on the heat and mass transfer during the ad- and desorption, are needed. In recent times, much research is being done in the area of materials development and materials testing for thermochemical energy storage applications. New material classes such as microporous aluminophosphates (AlPO), silicoaluminophophate (SAPO), metal organic frameworks (MOF) or composite materials of hygroscopic salts incorporated into a mesoporous carrier substance are subject of intense research ([7–11]). The positive developments in materials research promise novel and high performance storage materials in near future.

In this paper, a new binderless zeolite of type 13X (bead size 1.6–2.5 mm, Köstrolith© 13XBFK, [12]) of the company Chemiewerk Bad Köstritz, Germany, has been investigated in detail for thermochemical energy storage applications. This binderless zeolite is known to have a high adsorption capacity and a fast kinetics and is hence a promising storage material for thermochemical energy storage [35].

Information on the adsorption equilibrium and adsorption kinetics have been gained from isothermal measurements in a sorption analyzer and, on a larger scale at atmospheric pressure, by adsorption experiments in a fixed bed reactor. From these experimental investigations a numerical model of the fixed bed adsorption process with ambient air as carrier gas for heat and mass transfer has been derived. This numerical model can now be applied for the design process of a reactor for open thermochemical energy storage applications.

2. Materials and methods

In order to obtain information on the adsorption behavior of the binderless zeolite 13X, in particular on the adsorption equilibrium and adsorption kinetics, experimental investigations were performed. The measured data were then used to identify the parameters required to setup a numerical model of the adsorption process.

2.1. Experimental investigation

The sorption equilibrium of binderless zeolite 13X in a temperature range from 25 °C up to 200 °C was determined by gravimetric measurements of the water vapor adsorption isotherms. At higher temperatures of above 200 °C the hydrothermal stability of this zeolite is reduced [35]. The measurements were performed in a sorption analyzer IGA-002 (company Hiden Isochema). Prior to the measurements, the sample of approximately 50 mg of zeolite was dried in high vacuum (<10^{-5} mbar) at 350 °C. Adsorption isotherms (increase of the water vapor pressure at constant temperatures) and desorption isotherms (decrease of the water vapor pressure at constant temperatures) have been determined.

The macroscopic adsorption kinetics of the zeolites were investigated in a fixed bed reactor. A schematic drawing of the test rig which has been specially designed and built for these experimental investigations is depicted in Fig. 1. Central part of the setup is the fixed bed reactor (H = 140 mm, D = 50 mm) filled with beads of binderless zeolite 13X. The temperature profile inside the reactor is measured with thermocouples in three positions across the radius

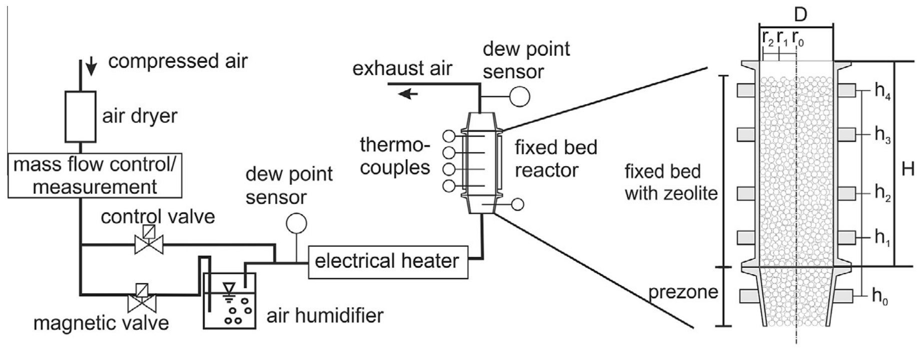

Fig. 1. Scheme of the test rig for fixed bed adsorption experiments (left) and a schematic detail drawing of the fixed bed reactor with approximately 250 ml of storage material.

(r_1 = 0 mm, r_2 = 12.5 mm, r_3 = 23 mm) in four parallel measuring planes (h_1 = 25 mm, h_2 = 55 mm, h_3 = 95 mm, h_4 = 125 mm). A "prezone" with inert particles of the same diameter as the zeolite beads is located directly upstream of the reactor. This "prezone" ensures a homogenous temperature and flow profile of the air.

The mass flow rate of the air is measured and controlled by a digital mass flow controller (Bronkhorst, El-Flow). The air is conditioned to the required adsorption humidity by an air humidifier and to the adsorption temperature by an electrically heated pipe. Upstream and downstream of the reactor the air humidity is measured with dew point sensors (EdgeTech Dew Masters DS2). The air inlet temperature is measured in the "prezone" (h_o = −25 mm) with a thermocouple. Before testing all thermocouples have been calibrated and adjusted. Hence, temperature differences can be measured with an accuracy of ±0.3 K, absolute temperature with an accuracy of ±1 K.

To minimize heat losses over the reactor wall the reactor is insulated with mineral wool. In addition, during desorption the reactor wall is heated up to the desorption temperature by electrical heating cables to guarantee a uniform temperature distribution over the reactor length and radius. Before each adsorption experiment the zeolite has been desorbed for 5 h in the fixed bed reactor at a temperature of 180 °C and a water vapor pressure of 10 mbar (equivalent to 40% relative humidity at 20 °C).

2.2. Governing equations of the numerical model

Both, the fluid flow in fixed bed reactors as well as the heat and mass transport during a chemical reaction have been studied in detail by many researchers (e.g. [13–19]). In the following the governing equations used in this study for the mathematical description of the water vapor adsorption of binderless zeolite 13X are summarized.

2.2.1. Adsorption equilibrium and heat of adsorption

The adsorption equilibrium X^* is approximated using the Dubinin-Astakhov (e.g. [20]) equation which is based on the micro-pore filling theory of Polanyi [21]. The adsorption volume W is defined by the maximum adsorption volume W_0, the characteristic energy E, the adsorption potential A and the heterogeneity factor of the micropore size distribution n:

$$W = W_0 \exp\left[-\left(\frac{A}{E}\right)^n\right] \tag{2}$$

with:

$$A = R_w T \ln\left(\frac{p_0}{p_w}\right) \tag{3}$$

In the Eq. (3) is p_0 the saturation pressure at the temperature T, p_w the partial water vapor pressure and R_w the specific gas constant of water vapor.

The adsorption equilibrium X^* is calculated from the adsorption volume according to the following equation:

$$X^* = \rho_{ads}(T) \cdot W \tag{4}$$

with the density of the adsorbed water ρ_{ads} [22]:

$$\rho_{ads}(T) = \frac{\rho_{20C}}{1 + \beta_{20C}(T - 293.15\,K)} \tag{5}$$

In Eq. (5) ρ_{20C} is the water density at 20 °C and β_{20C} the coefficient of thermal expansion of water vapor at 20 °C.

The heat of adsorption Δh_{ads} at constant water loading of the zeolite x_s is calculated with the van't Hoff equation:

$$-\Delta h_{ads} = R_w \left(\frac{\partial p_w}{\partial (1/T)}\right)_{x_s} \tag{6}$$

2.2.2. Momentum equation

The velocity profile $u_f(r)$ inside of the reactor is calculated with the extended Brinkman equation (e.g. [23,24,15]). The higher velocity of the fluid in near-wall regions is taken into account by a radial porosity profile (cf. Eq. (10)).

$$\frac{\partial p}{\partial z} = -f_1 u_f(r) - f_2 u_f(r)^2 + \frac{\eta_{eff}}{r}\frac{\partial}{\partial}\left(r\frac{\partial u_f(r)}{\partial r}\right) \tag{7}$$

The D'Arcy factor f_1 and the Forchheimer factor f_2 are strongly coupled to the bed porosity ε and are calculated according to [25]:

$$f_1 = 150\frac{[1-\epsilon(r)]^2}{\epsilon(r)^3}\frac{\eta_{eff}}{d_s^2}, \quad f_2 = 1.75\frac{[1-\epsilon(r)]^2}{\epsilon(r)^3}\frac{\rho_f}{d_s} \tag{8}$$

The effective viscosity η_{eff} is calculated with [26]:

$$\frac{\eta_{eff}}{\eta_f} = 2.0\exp(3.5 \cdot 10^{-3} Re_0), \quad Re_0 = \frac{u_{f,0} d_s \rho_f}{\eta_f} \tag{9}$$

In Eq. (7)–(9) is d_s the particle diameter, ρ_f the fluid density, η_f the fluid viscosity, Re_0 the particle Reynolds number, R_i the inner reactor radius and r and z the radial and axial directions.

The higher bed porosity ε in near wall regions is taken into account by a radial porosity profile which approaches a constant value of ε_∞ with increasing distance from the wall. The radial porosity profile is described with an exponential equation according to [26]:

$$\epsilon(r) = \epsilon_\infty\left(1 + 1.36\exp\left[-5.0\frac{R_i - r}{d_s}\right]\right) \tag{10}$$

2.2.3. Energy balance for fluid and solid phase

The heat and mass transport inside of the reactor is described with a quasi-homogenous model, which means that no difference in temperature is made between the solid and fluid phase. This model can be applied when the temperature gradient between the solid and fluid phase is small (c.f. [27,18]). This can reasonably be assumed for the fixed bed reactor under investigation due to the high surface area of the zeolite beads and the comparatively low released heat of the exothermic reaction.

With the assumption of a constant fluid density and a low water vapor content of the air flow the energy balance can be expressed with:

$$\left[\epsilon(r)\rho_f c_{p,f} + (1-\epsilon(r))\rho_s(c_{p,s} + x_s c_{p,ads})\right]\frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(\Lambda_r r\frac{\partial T}{\partial r}\right) + \frac{\partial}{\partial z}\Lambda_{ax}\frac{\partial T}{\partial z} - u_f(r)\rho_f c_{p,f}\frac{\partial T}{\partial z} + \rho_s(1-\epsilon(r))\frac{\partial}{\partial t}(\Delta h_{ads} x_s) \quad (11)$$

In Eq. (11) is T the quasi-homogenous fluid and solid temperature, $c_{p,f}$, $c_{p,s}$ and $c_{p,ads}$ the specific heat capacities of the fluid, zeolite particle and the adsorbed water, ρ_s the zeolite particle density, Δh_{ads} the heat of adsorption and x_s the water loading of the zeolite.

The effective axial and radial heat conductivity Λ_{ax} and Λ_r are described with the so-called Λ_r-model [24]. In near-wall regions the heat transport resistance increases leading to a steep temperature gradient near the wall. The corresponding equation for the effective radial and axial heat conductivity can be found e.g. in [24].

The specific heat capacity of the zeolite is divided into two terms: the specific heat capacity of the dry zeolite $c_{p,s}$ and the specific heat capacity of the adsorbed water $c_{p,ads}$. A constant value of the specific heat capacity of the dry zeolite has been assumed as there is no pronounced dependence on the temperature [36]. For the temperature dependent heat capacity of the adsorbed water an approximation suggested by Vucelic [28] is being used:

$$c_{p,ads}(T) = c_{p,min} + \int_{T_{min}}^{T} A\cdot\exp[-B(T-T_0)^2]dT \quad (12)$$

with the parameters (originally for a NaA-zeolite):

$$A = 3.762\text{e}-2;\quad B = 3.976\text{e}-4;\quad T_0 = 335\ \text{K};\quad c_{p,min} = 0.836\ \text{kJ/(kgK)}$$

2.2.4. Energy balance for the wall

In the experiments it has been observed that the reactor wall affects the dynamics of the adsorption process in the zeolite bed. This is mainly due to the comparatively high thermal conductivity and heat capacity of the stainless steel wall. To take these effects into account, an energy balance for the wall was included in the model.

$$\rho_R c_{p,R}\frac{\partial T_R}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(\lambda_R r\frac{\partial T_R}{\partial r}\right) + \frac{\partial}{\partial z}\lambda_R\frac{\partial T_R}{\partial z} \quad (14)$$

In Eq. (14) is T_R the reactor wall temperature, $c_{p,R}$ is the specific heat capacitiy, λ_R the heat conductivity and ρ_R the density of the reactor wall.

With the use of the Λ_r-model, there is no temperature gradient at the reactor wall/fixed bed boundary which means that that at the boundary $R = R_i$ the fixed bed temperature is equal to the wall temperature. The coupling of Eq. (11) and (14) is defined with:

$$T|_{r=R_i} = T_R|_{r=R_i} \quad (15)$$

Heat losses from the reactor wall to the ambient are expressed by an outward heat flux at the reactor wall/ambient boundary which is caused by a temperature difference between the reactor wall temperature T_R and the ambient temperature T_{amb}:

$$-\lambda_R\left[\frac{\partial T_R}{\partial r}\right]_{R_a} = h_R(T_{amb} - T_R|_{r=R_a}) \quad (16)$$

In Eq. (16) is h_R the heat transfer coefficient between the reactor wall and the ambient and R_a the outer diameter of the reactor wall.

2.2.5. Mass balance

The temporal change of the water content x_w in the fluid phase is calculated with:

$$\epsilon(r)\frac{\partial x_w}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(D_r r\frac{\partial x_w}{\partial r}\right) + \frac{\partial}{\partial z}D_{ax}\frac{\partial x_w}{\partial z} - u_f(r)\frac{\partial x_w}{\partial z} + \frac{\rho_s}{\rho_f}(1-\epsilon(r))\frac{\partial x_s}{\partial t} \quad (17)$$

The effective axial and radial dispersion coefficients D_{ax} and D_r are calculated according to [24].

The last term on the right hand side of Eq. (17) ($\partial x_s/\partial t$) is the temporal change of the water loading of the zeolite (adsorption rate). The adsorption rate is modeled by a linear driving force (LDF) approach as suggested for the first time by Glueckauf [29]. The LDF-model is widely used for adsorption modeling due to its mathematically simplicity and physical consistency [30–33]. The mass transfer resistance from the fluid phase into the micro-pores of the zeolite beads is reduced to an overall mass transfer resistance k_{LDF}. The driving force for adsorption or desorption is a concentration gradient between the adsorption equilibrium X^* at temperature T and water partial pressure p_w on the one hand and the actual loading of the zeolite x_s on the other hand.

$$\frac{\partial x_s}{\partial t} = k_{LDF}[X^*(T,p_w) - x_s] = 15\frac{\delta_{eff}}{(0.5d_s^2)}[X^*(T,p_w) - x_s] \quad (18)$$

The dominant mass transfer resistance for the water vapor adsorption of zeolite 13X is the macro-pore diffusion resistance D_M [31]. The macro-pore diffusion is a superposition of molecular diffusion and gas diffusion. The corresponding equations can be found e.g. in [34]. The effective diffusion coefficient δ_{eff} is then calculated with Eq. (19):

$$\delta_{eff} = \frac{D_M}{\mu}\frac{\epsilon_s}{\rho_s R_w T}\frac{1}{\partial X^*/\partial p_w} \quad (19)$$

In Eq. (19) is ε_s the particle porosity and μ the tortuosity factor of the macropore network.

3. Results and discussion

3.1. Determination of the adsorption equilibrium and heat of adsorption

The left Fig. 2 shows the measured adsorption isotherms of binderless zeolite 13X over a wide temperature and partial water vapor pressure range. In addition, the adsorption isotherms calculated with Eqs. (2)–(4) are depicted. The corresponding values of the parameters fitted with a nonlinear least-square fit to the measured data are:

$$W_0 = 341.03\ \text{ml/g};\quad E = 1192.3\ \text{kJ/kg};\quad n = 1.55$$

A good agreement between the experimental and calculated data is obtained and confirms the applicability of the Dubinin–Astakhov equation for the description of the adsorption equilibrium of the zeolite under investigation.

The right diagram of Fig. 2 shows the heat of adsorption of the zeolite calculated with Eq. (6) and from calorimetric measurements performed by Bering et al. [21] for powdered zeolite 13X. It can be seen that Eq. (6) provides a good approximation for the calculation of the heat of adsorption.

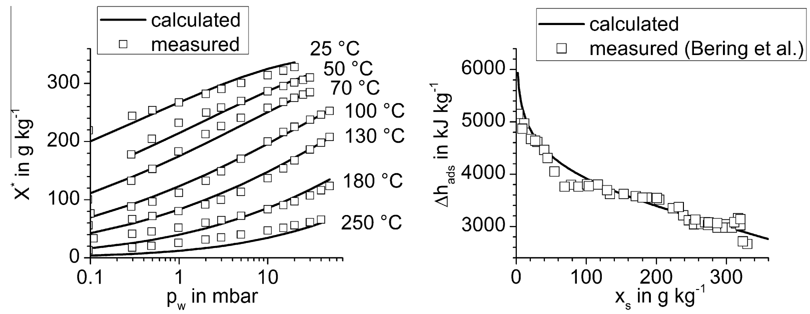

Fig. 2. Measured (solid squares) and calculated (line + open squares) water vapor adsorption isotherms of binderless zeolite 13X (left) and heat of adsorption as a function of the water loading calculated according to Eq. (6) and from calorimetric measurements [21], (right).

3.2. Determination of the thermal conductivity, specific heat capacity and adsorption kinetics of binderless zeolite 13X

In a first step, experimental investigations have been performed in the fixed bed reactor without adsorption or desorption. The aim of these experiments was the determination of the thermal conductivity and specific heat capacity of binderless zeolite 13X without the influence of the adsorption/desorption process. In addition the heat transfer coefficient describing the heat losses via the reactor wall has been determined. In a next step further experiments have been performed to investigate the adsorption kinetics.

For all experiments, the reactor has been filled with zeolite up to the forth measuring plane (h_4 = 125 mm), corresponding to 170 g of dry zeolite. The bulk density of zeolite is calculated to ρ_{eff} = 690 kg/m^3. Assuming a porosity of the fixed bed of ε = 0.4 the solid particle density is calculated to ρ_s = 1150 kg/m^3.

For the experiments without adsorption the fixed bed reactor has been heated up by the air flow in steps from 30 °C to 130 °C and then cooled down. Each temperature step (30 °C, 50 °C, 70 °C, 100 °C, and 130 °C) has been kept for four hours. Adsorption or desorption has been suspended during heating and cooling by using a dry bed of zeolite and an air flow with a very low water content of <0.5 g/kg.

With the numerical model the heating and cooling experiments were simulated using the measured air inlet temperature and air humidity as input values. The values of the thermal conductivity and specific heat capacity used for the numerical model have been varied (variation of the thermal conductivity from 0.1 W/(m K) to 0.6 W/(m K), variation of the specific heat capacity from 0.8 kJ/(kg K) to 1.0 kJ/(kg K) and variation of the heat transfer coefficient from 0 to 3 W/(m^2 K)) until good agreement between the measured and the calculated temperature profile in the fixed bed reactor was obtained.

It has been found out that the numerical model is very sensitive to variations in the thermal conductivity of the zeolite. Especially the radial temperature profile is strongly correlated to this value.

The overall adsorption kinetics of the binderless zeolite 13X have then been studied by further experimental investigations in the fixed bed reactor. The experiments have been performed at different air inlet temperatures (30 °C to 50 °C) and at different air humidities (5 to 30 mbar).

Fig. 3 shows the temperature in the radial center of the fixed bed at position h_4 (end of the fixed bed) for different adsorption experiments performed at an air flow inlet temperature of 30 °C. The maximum temperature increase in the fixed bed reactor is almost proportional to the humidity of the air flow: 11 K at a water content of the air of 3.1 g/kg (5 mbar; Exp 1), 38 K at a water content of 9.5 g/kg (15 mbar; Exp 2) and 78 K at a water content of 19.2 g/kg (30 mbar; Exp 2).

The partial water vapor pressure of the airflow at the reactor inlet and outlet is depicted in gray lines. Over a long period of time the water vapor pressure at the reactor outlet is below the resolution of the dew point sensors (<1 mbar or <0.5 g/kg respectively). Almost the complete water content of the air is adsorbed by the zeolite which shows the high affinity of zeolite towards water vapor. The stepwise increase and decrease in temperature and the stepwise increase in the water content of the outflowing air further indicates the fast reaction kinetics of the binderless zeolite 13X.

The experimental data of the adsorption process together with the developed numerical model have been used to determine the model parameters describing the adsorption kinetics (macro-pore diameter, tortuosity). The same method as described above has been used for the parameter estimation. The macro-pore diameter has been varied from $d_{p,macro}$ = 30 to 400 nm, the tortuosity from μ = 1 to 10. The particle porosity has been set constant to ε_s = 0.6.

Good agreement between experiment and simulation has been obtained for the following values of the parameters:

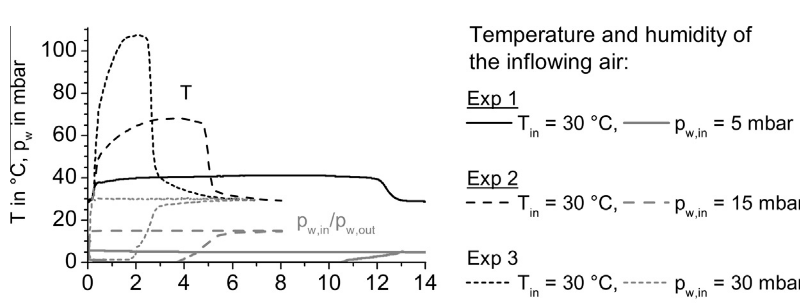

Fig. 3. Temperature at position h_4 = 125 mm and water vapor pressure of the airflow (inlet/outlet) during adsorption experiments by varying the humidity of the air.

Parameter	Symbol	Value
thermal conductivity of the zeolite	λ_s	0.4 W/(m K)
specific heat capacity of the zeolite	$c_{p,s}$	0.88 kJ/(kg K)
macro-pore diameter of the zeolite	$d_{p,macro}$	300 nm
tortuosity of the zeolite	μ	4.0

The heat transfer coefficient has been determined to $h_R = 1.4$ W/(m^2 K).

It should be noted that there is a strong correlation between the parameters determined. Especially the heat transfer coefficient and the thermal heat conductivity of the zeolite are strongly linked. Hence, a transfer of these values to other applications e.g. to closed sorption systems should be carefully checked.

Figs. 4 and 5 exemplarily show the measured temperature (dotted lines) and the temperature calculated with the numerical model (solid lines) at different axial ($h_1 = 25$ mm, $h_2 = 55$ mm, $h_3 = 95$ mm, $h_4 = 125$ mm) and radial ($r_0 = 0$ mm, $r_1 = 12.5$ mm, $r_2 = 23$ mm) positions in the fixed bed reactor. Fig. 4 shows an adsorption experiment which has been performed with an air inlet temperature of 30 °C and a water vapor pressure of 15 mbar; Fig. 5 shows an adsorption experiment with an air inlet temperature of 30 °C and 5 mbar water vapor pressure (left) and at 50 °C and 15 mbar water vapor pressure (right).

From the temperatures measured at the longitudinally distributed sensors, the adsorption front moving in axial direction through the reactor is clearly visible. At the beginning of the adsorption the heat released is used to heat up the storage material to the adsorption temperature. After that, during almost the complete adsorption a constant temperature is maintained at the outlet of the fixed bed reactor (at position h_4). After a certain time of adsorption, the temperature first starts to decrease at position h_1, then at h_2, h_3 and h_4.

The radial temperature distribution depicted in Fig. 4 illustrates the influence of the reactor wall on the fixed bed temperature. In the near-wall region (radial position r_3) the maximum temperature is significantly lower than in the central part of the reactor (radial position $r_{1,}$ r_2). This is caused by heat losses over the reactor wall. Furthermore, the heating and cooling of the zeolite in the near-wall region is slower due to the comparatively high heat capacity of the wall which is by a factor of about 2.5 higher compared to the heat capacity of the fixed bed of zeolites.

The good agreement of numerical and experimental results shows the importance of including the reactor wall into the simulation model.

The experimental results also show that the temperature decrease in the near-wall regions starts earlier than in the central part of the reactor. This can be explained by the higher porosity and the higher fluid velocity (due to the higher porosity) in the near-wall region which lead to a faster completion of the adsorption process. This explanation is confirmed by the numerical results which show that in near wall-regions the water uptake by the material is faster completed than in the center of the reactor.

By applying the radial porosity profile in both, the momentum and energy equation, this phenomenon can be displayed also numerically.

The temperature distribution in the fixed bed reactor calculated with the developed numerical model reflects the measured temperature distribution very good. The early decrease in temperature as well as the slower heating and cooling of the zeolite in near-wall regions compared to the center of the reactor is described very well by the numerical model. For all experiments

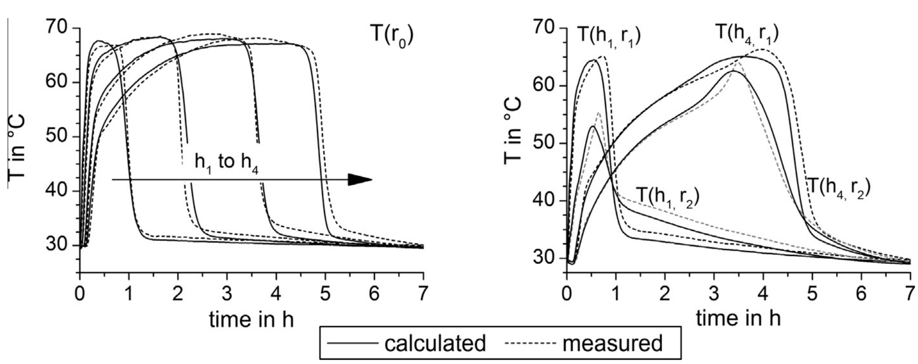

Fig. 4. Measured (dotted lines) and calculated (solid lines) temperatures during water vapor adsorption with an air inlet temperature of 30 °C and a water vapor pressure of 15 mbar. Left: Temperatures at the four horizontal positions in the center of the fixed bed reactor ($r_0 = 0$ mm) Right: Temperatures at position h_1 and h_4 at $r_2 = 12.5$ mm and $r_3 = 23$ mm.

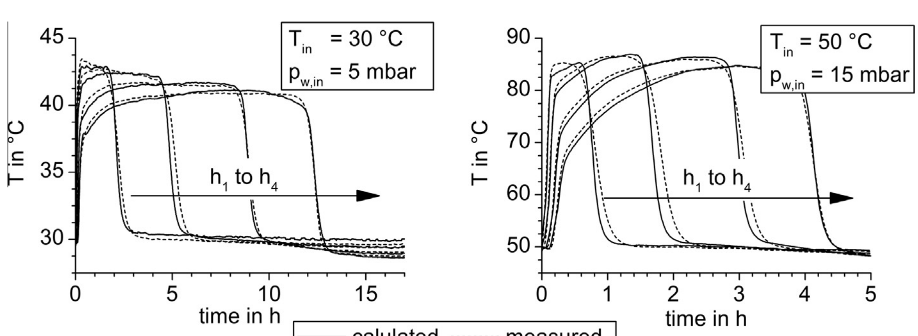

Fig. 5. Measured (dotted lines) and calculated (solid lines) temperatures in the four axial measuring planes in the center of the fixed bed ($r_0 = 0$); Left: air flow inlet temperature 30 °C, partial water vapor pressure 5 mbar Right: air flow inlet temperature 50 °C, partial water vapor pressure 15 mbar.

and for all temperature sensors a good agreement between the calculated and the measured temperatures is obtained.

4. Conclusion

Experimental investigations have been performed to determine the water vapor adsorption behavior of binderless zeolite 13X. A numerical model of a fixed bed adsorption process with humid air as heat and mass transport medium has been set up. Experimental data have been used to determine the model parameters and to validate the numerical model.

The experimental results clearly show the influence of the wall on the temperature profile in the fixed bed reactor. The higher bed porosity in near wall regions leads to a faster completion of the adsorption; the additional heat capacity of the wall reduces the cooling rate of the zeolite in near wall regions.

In the numerical model this has been considered by a radial porosity profile and by an additional energy equation for the wall. By this, the thermal effects in the fixed bed reactor caused by the wall could be approximated very well by the numerical model.

The numerical model allows an in-depth understanding and a visualization of the heat and mass transport during adsorption. With this validated numerical model it is now possible to investigate the dynamic behavior of the adsorption process in different applications and for different boundary conditions. In particular, the numerical model provides an excellent basis for the development process of a reactor design for open thermochemical energy storage and allows targeted designing of the reactor.

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